Lab-Manual – IAM – CLI

Objective of this LAB

Create an EC2 Instance from the CLI.

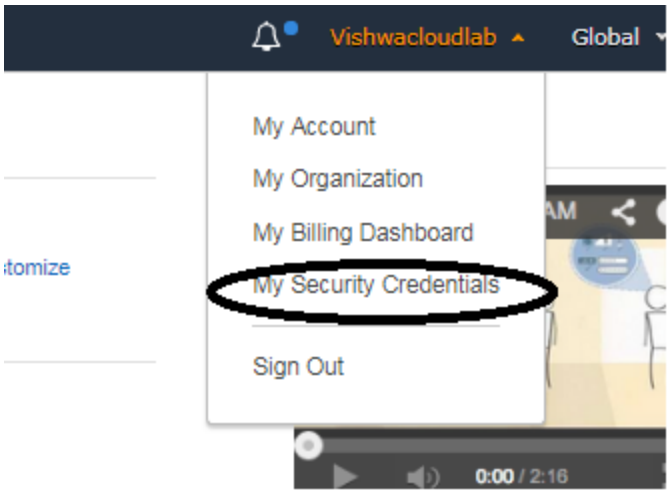
Note:-- Key, VPC, Subnet and SG should have been created before we proceed further

Steps:

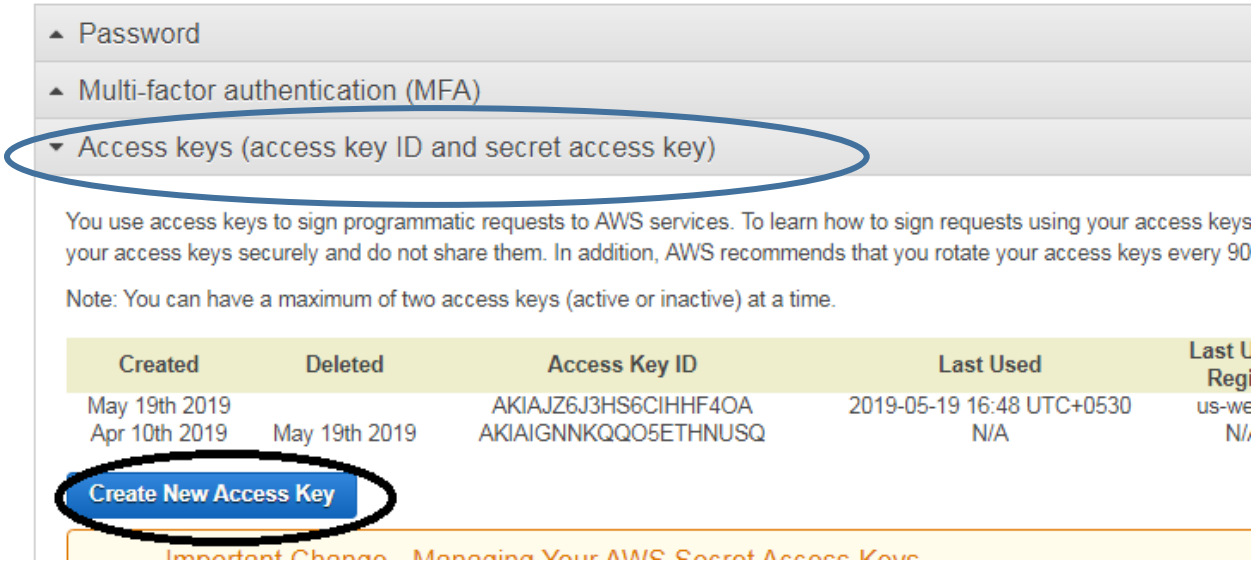
1. Enable CLI Access for a root user.
2. Login to AWS via CLi and verify the login
3. Create a json template for creating an EC2 instance.
4. Run the CLI for the above json

1. Enable CLI Access for a root user.

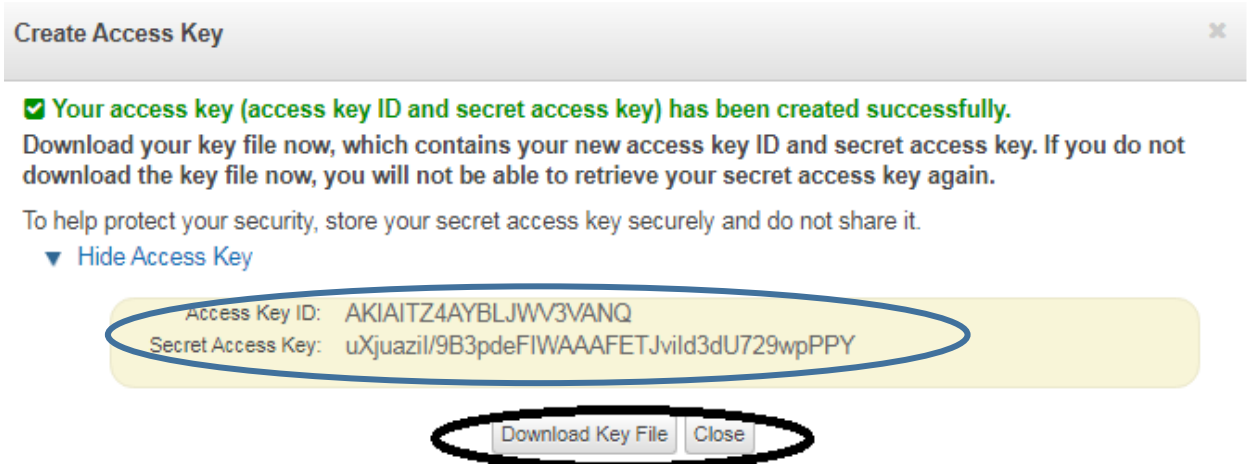
   I.



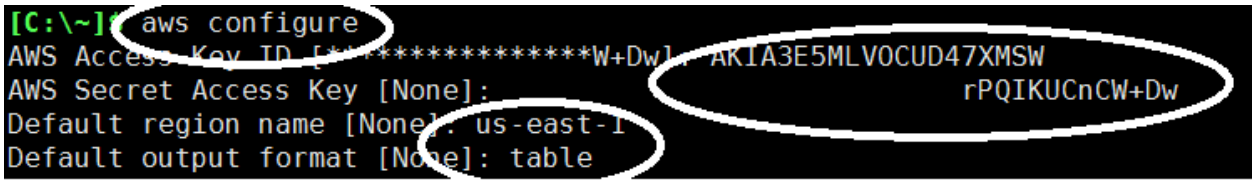
   II.



III. Once the key is created, copy the "Secret access key" as it would **NOT** be **SAVED** in AWS



2. Login to AWS via CLi and verify the login



 # aws configure

Execute → $ aws ec2 describe-vpcs

If you get an TABLE output means, your credentials are working fine.

3. Create a json template for creating an EC2 instance.

Copy the below content with the same format as <filename>.json

Note: -- If the copy does not work fine, the "json" is also uploaded in the drive in the same folder.

```
{
    "ImageId": "ami-b70554c8",
    "InstanceType": "t2.micro",
    "KeyName": "b16-key1",
    "MaxCount": 1,
    "MinCount": 1,
    "Monitoring": {
        "Enabled": true
    },
    "SecurityGroupIds": [
        "sg-63c93c2e"
    ],
    "SubnetId": "subnet-d382c599",
    "UserData": "",
    "DisableApiTermination": true,
    "DryRun": false,
    "EbsOptimized": false,
    "IamInstanceProfile": {
        "Arn": "",
        "Name": ""
    },
    "InstanceInitiatedShutdownBehavior": "stop",
    "NetworkInterfaces": [
        {
            "AssociatePublicIpAddress": true,
            "DeleteOnTermination": true,
            "Description": "linuxNic1",
            "DeviceIndex": 0,
            "Groups": [
                ""
            ],
            "NetworkInterfaceId": "",
            "PrivateIpAddress": "192.168.1.101"
        }
    ]
}
```

4. Run the CLI for the above json with “dry-run” enabled.

$ aws ec2 run-instances --cli-input-json file://Create-instance-01.json - -dry-run

Make sure you have changed the directory to the json file folder as example below.

```
[D:\Tech-docs\AWS\Re-Ordered-Documents\12-AWS-IAM\Sample-json-CLI]$ dir
 Volume in drive D has no label.
 Volume Serial Number is 8E46-329B

 Directory of D:\Tech-docs\AWS\Re-Ordered-Documents\12-AWS-IAM\Sample-json-CLI

05/17/2019  05:40 PM    <DIR>          .
05/17/2019  05:40 PM    <DIR>          ..
05/19/2019  04:37 PM               599 Create-instance-01.json
03/05/2019  08:50 AM             1,944 Create-instance-01.txt
03/05/2019  09:12 AM             1,273 Create-instance-02.json
               3 File(s)          3,816 bytes
               2 Dir(s)  29,931,687,936 bytes free

[D:\Tech-docs\AWS\Re-Ordered-Documents\12-AWS-IAM\Sample-json-CLI]$
```

Output:

```
[D:\Tech-docs\AWS\Re-Ordered-Documents\12-AWS-IAM\Sample-json-CLI]$ aws ec2 run-instances --cli-input-json file://Create-instar
ce-01.json --dry-run

An error occurred (DryRunOperation) when calling the RunInstances operation: Request would have succeeded, but DryRun flag is s
et.
```

This shows that “Request would have Succeeded, but DryRun flag is set”

This message confirms that the JSON file format seems to be fine and you can execute the file.

Lets now run the same command without the DRYRUN

5. Run the CLI for the above json with “dry-run” diabled.

$ aws ec2 run-instances --cli-input-json file://Create-instance-01.json

Output:

```
-----------------------------------------------------------------------------
|                              RunInstances                                 |
+------------------------------+----------------------------------------+
|  OwnerId                     |  558457644648                          |
|  ReservationId               |  r-087073d3f620b42dd                   |
+------------------------------+----------------------------------------+
||                                Instances                                ||
|+------------------------+-----------------------------------------------+|
||  AmiLaunchIndex        |  0                                            ||
||  Architecture          |  x86_64                                       ||
||  ClientToken           |                                               ||
||  EbsOptimized          |  False                                        ||
||  Hypervisor            |  xen                                          ||
||  ImageId               |  ami-061392db613a6357b                        ||
||  InstanceId            |  i-03f114e099e70cfb0                          ||
||  InstanceType          |  t2.micro                                     ||
||  KeyName               |  Key-VCL-Ore-01                               ||
||  LaunchTime            |  2019-05-19T11:07:33.000Z                     ||
||  PrivateDnsName        |  ip-172-31-52-104.us-west-2.compute.internal  ||
||  PrivateIpAddress      |  172.31.52.104                                ||
||  PublicDnsName         |                                               ||
||  RootDeviceName        |  /dev/xvda                                    ||
```